MATLAB Home - MATLAB

HOME | PLOTS | APPS | FIGURE | FORMAT | TOOLS

Search (Ctrl+Shift+Space) — Deveswar

Save As | Show Code | FILE

Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid

LABELS AND ANNOTATIONS

/ › MATLAB Drive ›

Editor

exp1dsp.m × | exp2dsp.m × | exp3dsp.m × | exp4dsp.m × | exp5dsp.m ×

/MATLAB Drive/exp5dsp.m

```
clc; clear; close all;

%% IIR Filter Coefficients
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

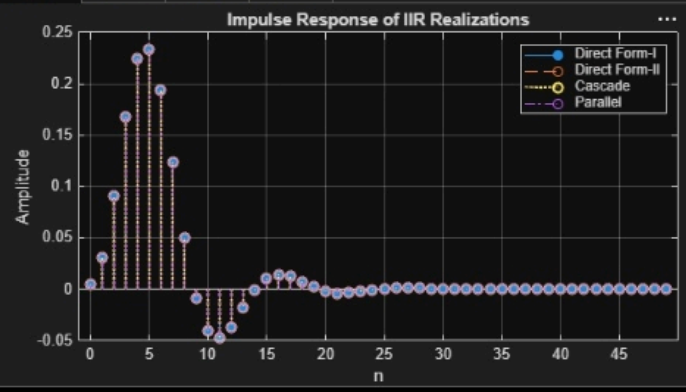
N = 50;
x = [1 zeros(1,N-1)]; % Unit impulse
n = 0:N-1;

%% 1. Direct Form-I and Direct Form-II
y1 = filter(b,a,x); % Direct Form-I
y2 = filter(b,a,x); % Direct Form-II

%% 2. Cascade Realization
[sos,g] = tf2sos(b,a);
yc = g*x;
```

Figures

Figure 1 × | Figure 2 × | Figure 3 × | Figure 4 ×



Command Window

Comparison Table:

Structure	Maximum_Error	Deviation_percent
"DF-I"	0	0
"DF-II"	0	0
"Cascade"	1.9221e-15	8.242e-13

31°C Sunny | Search | ENG IN | 10:44 13-10-2026

MATLAB Home - MATLAB

HOME | PLOTS | APPS | FIGURE | FORMAT | TOOLS

Search (Ctrl+Shift+Space)

Deveswar

Save As | Show Code | FILE

Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid

LABELS AND ANNOTATIONS

/ › MATLAB Drive ›

Editor

exp1dsp.m × | exp2dsp.m × | exp3dsp.m × | exp4dsp.m × | exp5dsp.m ×

/MATLAB Drive/exp5dsp.m

```
clc; clear; close all;

%% IIR Filter Coefficients
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

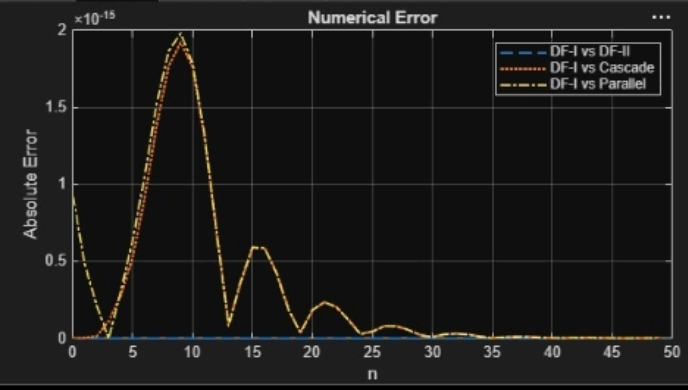
N = 50;
x = [1 zeros(1,N-1)]; % Unit impulse
n = 0:N-1;

%% 1. Direct Form-I and Direct Form-II
y1 = filter(b,a,x); % Direct Form-I
y2 = filter(b,a,x); % Direct Form-II

%% 2. Cascade Realization
[sos,g] = tf2sos(b,a);
yc = g*x;
```

Figures

Figure 1 × | Figure 2 × | Figure 3 × | Figure 4 ×



Command Window

Comparison Table:

Structure	Maximum_Error	Deviation_percent
"DF-I"	0	0
"DF-II"	0	0
"Cascade"	1.9221e-15	8.242e-13

31°C Sunny

Search

ENG IN

10:45 13-10-2026

MATLAB Home - MATLAB

HOME | PLOTS | APPS | FIGURE | FORMAT | TOOLS

Search (Ctrl+Shift+Space)

Deveswar

Save As | Show Code | FILE

Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid

LABELS AND ANNOTATIONS

/ › MATLAB Drive ›

Editor

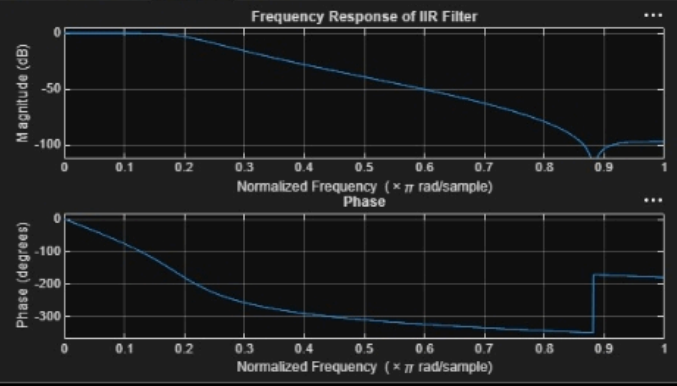
exp1dsp.m × | exp2dsp.m × | exp3dsp.m × | exp4dsp.m × | exp5dsp.m ×

/MATLAB Drive/exp5dsp.m

```
1   clc; clear; close all;
2
3   %% IIR Filter Coefficients
4   b = [0.0048 0.0193 0.0289 0.0193 0.0048];
5   a = [1 -2.3695 2.3140 -1.0547 0.1874];
6
7   N = 50;
8   x = [1 zeros(1,N-1)]; % Unit impulse
9   n = 0:N-1;
10
11  %% 1. Direct Form-I and Direct Form-II
12  y1 = filter(b,a,x); % Direct Form-I
13  y2 = filter(b,a,x); % Direct Form-II
14
15  %% 2. Cascade Realization
16  [sos,g] = tf2sos(b,a);
17  yc = g*x;
```

Figures

Figure 1 × | Figure 2 × | Figure 3 × | Figure 4 ×



Command Window

Comparison Table:

Structure	Maximum_Error	Deviation_percent
"DF-I"	0	0
"DF-II"	0	0
"Cascade"	1.9221e-15	8.242e-13

31°C Sunny

Search

ENG IN

10:45 13-10-2026

MATLAB Home - MATLAB

HOME PLOTS APPS FIGURE FORMAT TOOLS

Search (Ctrl+Shift+Space)

Deveswar

Save As | Show Code | FILE

Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid

LABELS AND ANNOTATIONS

/ › MATLAB Drive ›

Editor

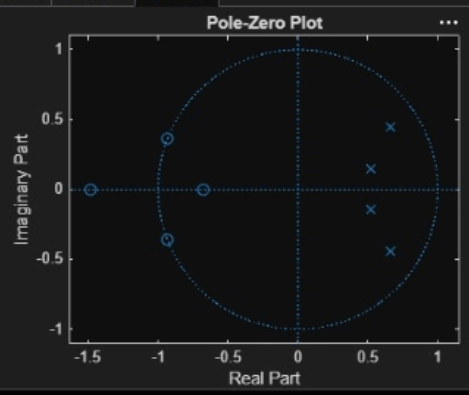
exp1dsp.m | exp2dsp.m | exp3dsp.m | exp4dsp.m | exp5dsp.m

/MATLAB Drive/exp5dsp.m

```
1   clc; clear; close all;
2
3   %% IIR Filter Coefficients
4   b = [0.0048 0.0193 0.0289 0.0193 0.0048];
5   a = [1 -2.3695 2.3140 -1.0547 0.1874];
6
7   N = 50;
8   x = [1 zeros(1,N-1)]; % Unit impulse
9   n = 0:N-1;
10
11  %% 1. Direct Form-I and Direct Form-II
12  y1 = filter(b,a,x); % Direct Form-I
13  y2 = filter(b,a,x); % Direct Form-II
14
15  %% 2. Cascade Realization
16  [sos,g] = tf2sos(b,a);
17  yc = g*x;
```

Figures

Figure 1 | Figure 2 | Figure 3 | Figure 4



Command Window

```
Comparison Table:
    Structure      Maximum_Error      Deviation_percent
    _________      _____________      _________________

    "DF-I"                     0                      0
```